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Course : Programming in java

Course code : CSA0974

[1] QID: 37

Question : Find the Mth maximum number and Nth minimum number in an array and then find the sum of it and difference of it.

Sample Input:

Array of elements = {14, 16, 87, 36, 25, 89, 34}

M = 1

N = 3

Sample Output:

1stMaximum Number = 89

3rdMinimum Number = 25

Sum = 114

Difference = 64

Testcases:

1. {16, 16, 16 16, 16}, M = 0, N = 1
2. {0, 0, 0, 0}, M = 1, N = 2
3. {-12, -78, -35, -42, -85}, M = 3 , N = 3
4. {15, 19, 34, 56, 12}, M = 6 , N = 3
5. {85, 45, 65, 75, 95}, M = 5 , N = 7

Program :

```
import java.io.*;
import java.util.*;
 class Find
{
 public static void main (String args[])
 {
 Scanner s=new Scanner(System.in);
 int k=s.nextInt();
int[] arr=new int[k];
for(int i=0;i
```

Input :

7

14 16 87 36 25 89 34

1

3

Output :

1st max=89

3rd min=25

sum=114

diff=64

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[2] QID: 42

Question : Write a program to print the first n perfect numbers. (Hint Perfect number means a positive integer that is equal to the sum of its proper divisors)

Sample Input:

N = 3

Sample Output:

First 3 perfect numbers are: 6 , 28 , 496

Testcases:

1. N = 0

2. N = 5

3. N = -2

4. N = -5

5. N = 0.2

Program :

```
import java.io.*;

import java.util.*;

class Perfect

{

public static void main (String args[])

{

int n=3;

Scanner s=new Scanner(System.in);
```

```
n=s.nextInt();
if(n==0)
{System.out.println("not perfect");
return;}
int count=0,num=2;
while(count
```

Input :

3

Output :

6

28

496

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[3] QID: 39

Question : Write a program using choice to check

Case 1: Given string is palindrome or not

Case 2: Given number is palindrome or not

Sample Input:

Case = 1

String = MADAM

Sample Output:

Palindrome

Testcases:

1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321

Program :

```
import java.io.*;
import java.util.*;
class Palindrome
{
public static void main (String args[])
{
Scanner sc=new Scanner(System.in);
int ch=sc.nextInt();
switch(ch){
case 1:
String str=sc.next();
String rev="";
for(int i=str.length()-1;i>=0;i--)
{
rev+=str.charAt(i);
}
```

```
if(str.equals(rev))
System.out.println("it is palindrome");
else
System.out.println("it is not palindrome");
 break;


 case 2:
 int num=sc.nextInt();
 int tem=num;
 int revNum=0;
 while(num>0){
 revNum=revNum*10+num%10;
 num/=10;
 }
 if(tem==revNum)
 System.out.println("it is palindrome");
 else
 System.out.println("it is not palindrome");
 break;
 default:
 System.out.println("invalid");
 }
 sc.close();
 }
```

}

Input :

2

121

Output :

it is palindrome

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